Fibonacci Series

```
def fibonacci(n):
    if n<=0:
        return []
    elif n==1:
        return [0]
    series = [0,1]
    for _ in range(2,n):
        series.append(series[-1] + series[-2])
    return series
```

(or)

```
for i in range(2,n):
    series.append(list[i-1] + list[i-2])
```

Exception

```
try:
except:
else
finally
```

to mention to handle for proper exception use

can use multiple except

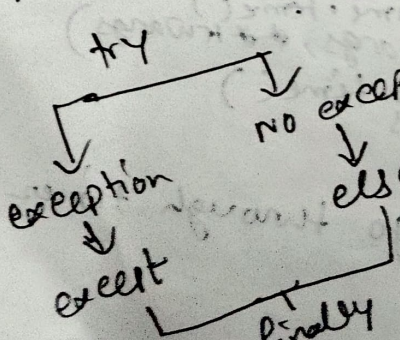
```
try:
except ValueError:
```

↓ it'll handle the error, only if it's ValueError.

(or) except(type, type, ...)

else only executes if except block don't work.

finally → will execute definitely,



to get actual error

```
try:
    res = 10/0
except ZeroDivisionError as e:
    print(e)
```

o/p division by zero

Enumerate → is a built-in python fun^n that lets you loop through a sequence while getting both index & value

for i in range(len(names)):
    print(i, names[i])

for i, name in enumerate(name):
    print(i, name).

produces pairs → (idx, value).

Syntax enumerate(iterable, (itr, def_idx)) ↳ optional.

(#) Find a target & it's index.

```
n = [10,20,30,40,50]
for i,val in enumerate(n):
    if (val == target):
        print(i)
```

(#) x = {"masala", "Geiger", "lemon"} → tuple.

```
y = enumerate(x)
list(y)
```

```
f = open('test.py','w')
f.write(' ')
f.close()
```

→ it'll create a new file with following name.
→ to add text to file.

(#) or

```
with open('youtube.txt','w') as file:
    file.write("chai aur python")
```

(#)

```
if __name__ == '__main__':
    main()
```

(#) JSON.

```
import json
json.load(file)
.dump(kya likhna hai, kaha)
.dump(videos, file)
```

→ String converted into JSON.

SQLite3

```
import sqlite3
conn = sqlite3.connect("tutorial.db")
```

↳ returns a Connection object.

(#) we need to create a cursor.

```
cursor = conn.cursor()
```

~~cursor.execute("")~~

conn.close() → to close the connection.

cursor.executed("""
SQL commands only
""")

SQLite Datatypes

Integer, real, text, NULL

real → decimal

ex

```
cursor.execute(
    "Insert into users(name, age)
     values(?, ?)", ("Ray", 25))
```

conn.commit() → to save the progress.

to insert multiple param

```
cursor.executemany("""
    Insert into users(name, age)
    values(?, ?)""",
    [('Rajib', 25),
     ('Sid', 24),
     ...])
```

to use multiple statements / commands use

.executescript(

,

Fetching data

cursor.executed("select * from users")

—this only executes the query, but we need to fetch the res.

fetchone() → returns one row only

① —

② row = cursor.fetchone()
print(row)

fetchall()

① —

② rows = cursor.fetchall()
for row in rows:
print(row).

fetchmany(2) → how many rows

VVI for a single param

use (age,), is mandatory as it is a python tuple

Notebook

notebook is a special file (.ipynb) that lets you add notes, eqn, images, charts etc

Jupyter notebooks → using anaconda